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#RBT23CB059 MANALI GHULE
# EXPERIMENT: POS TAGGING AND SYNTAX ANALYSIS

# Step 1: Install required libraries
!pip install nltk spacy

!python -m spacy download en_core_web_sm

# Step 2: Import libraries
import nltk
import spacy

from nltk import RegexpParser
from IPython.display import display
from spacy import displacy

# Step 3: Download NLTK resources
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('averaged_perceptron_tagger_eng')

# Step 4: Load spaCy model
nlp = spacy.load("en_core_web_sm")

# Step 5: Input sentences
sentences = [
    "The boy is playing cricket.",
    "She reads a book.",
    "The dog runs in the park."
]

# Step 6: POS Tagging
print("========== POS TAGGING ==========")

for sentence in sentences:

    words = nltk.word_tokenize(sentence)

    tagged_words = nltk.pos_tag(words)

    print("\nSentence:", sentence)
    print("POS Tags:")

    for word, tag in tagged_words:
        print(word, ":", tag)


# Step 7: Syntax Analysis using Dependency Parsing
print("\n========== SYNTAX ANALYSIS ==========")

for sentence in sentences:

    doc = nlp(sentence)

    print("\nSentence:", sentence)

    for token in doc:
        print(
            "Word:", token.text,
            "| POS:", token.pos_,
            "| Dependency:", token.dep_,
            "| Head:", token.head.text
        )


# Step 8: Phrase Structure Parsing
print("\n========== PHRASE STRUCTURE PARSING ==========")

grammar = r"""
    NP: {<DT>?<JJ>*<NN.*>+}
    VP: {<VB.*><NP>?}
    PP: {<IN><NP>}
"""

parser = RegexpParser(grammar)
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for sentence in sentences:

    words = nltk.word_tokenize(sentence)

    tagged = nltk.pos_tag(words)

    tree = parser.parse(tagged)

    print("\nSentence:", sentence)

    tree.pretty_print()


# Step 9: Dependency Graph Visualization
print("\n========== DEPENDENCY GRAPH ==========")

for sentence in sentences:

    doc = nlp(sentence)

    print("\nDependency graph for:", sentence)

    display(displacy.render(
        doc,
        style="dep",
        jupyter=True
    ))


# Step 10: Final Result
print("\n================================")
print("POS TAGGING AND SYNTAX ANALYSIS")
print("IMPLEMENTED SUCCESSFULLY")
print("================================")
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